

UNIT 2

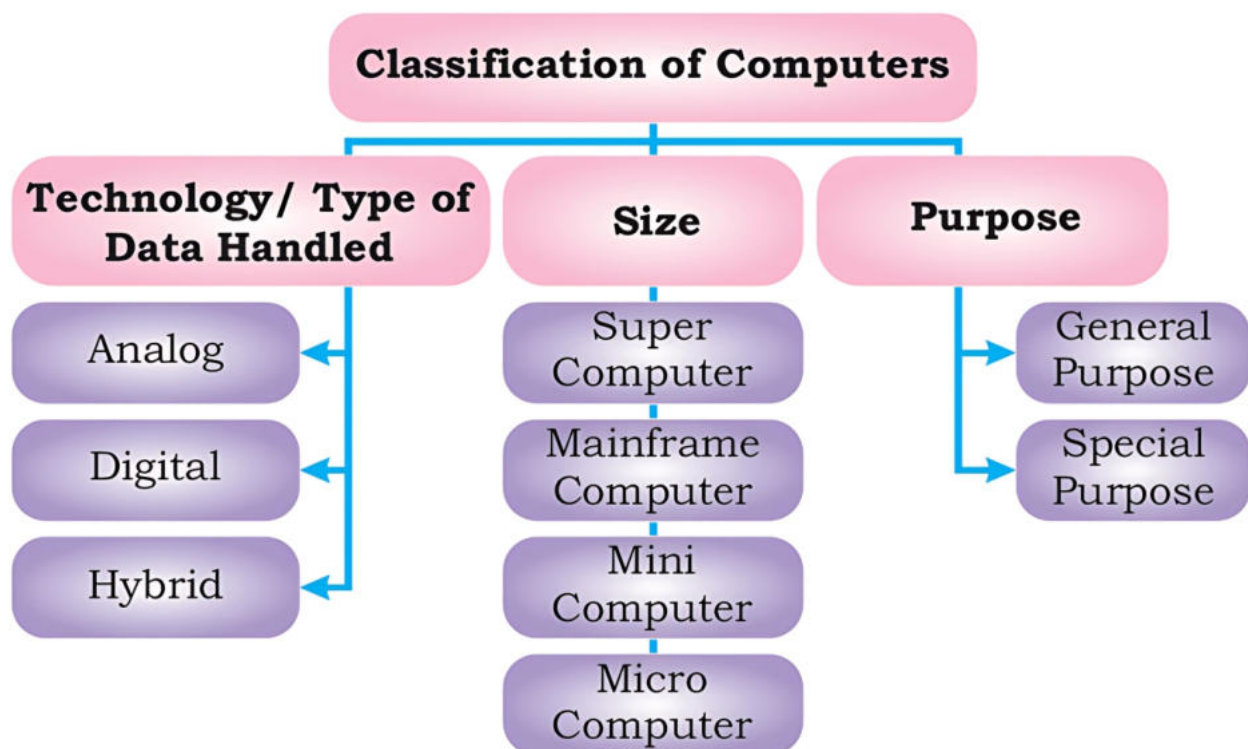
CLASSIFICATION OF COMPUTERS



Learning Objectives

1. Understand how computers are classified based on the type of data they handle.
2. Differentiate between analog, digital, and hybrid computers.
3. Identify the purposes of different types of digital computers.
4. Recognize the classification of digital computers based on size and performance.
5. Provide examples of applications for each category of computer.

Computers are versatile machines used in various fields, but not all computers function the same way. Depending on how they handle data, their purpose, and their size and performance, computers are classified into multiple categories. Understanding these classifications helps in selecting the right computer for specific tasks.



Types of Computers According to the Data They Handle

1. Analog Computers

Analog computers process data represented in the form of continuously varying physical quantities such as pressure, temperature, voltage, and current. They measure things and are used in applications where variables change continuously.

Applications:

- Speedometer of a car
- Gasoline pumps

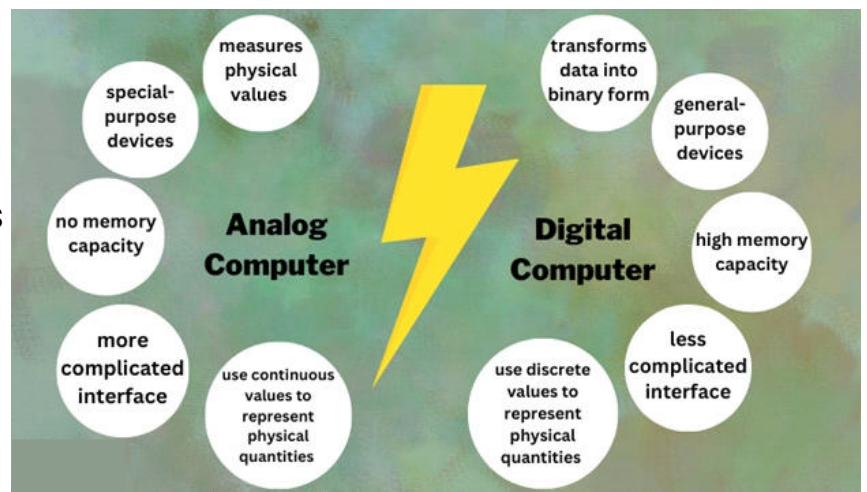


2. Digital Computers

Digital computers operate on discrete data, such as numbers, and solve problems by performing arithmetic and logical operations based on stored programs. They count things and are widely used in various fields.

Applications:

- Home
- Educational institutions
- Offices
- Scientific research
- Business

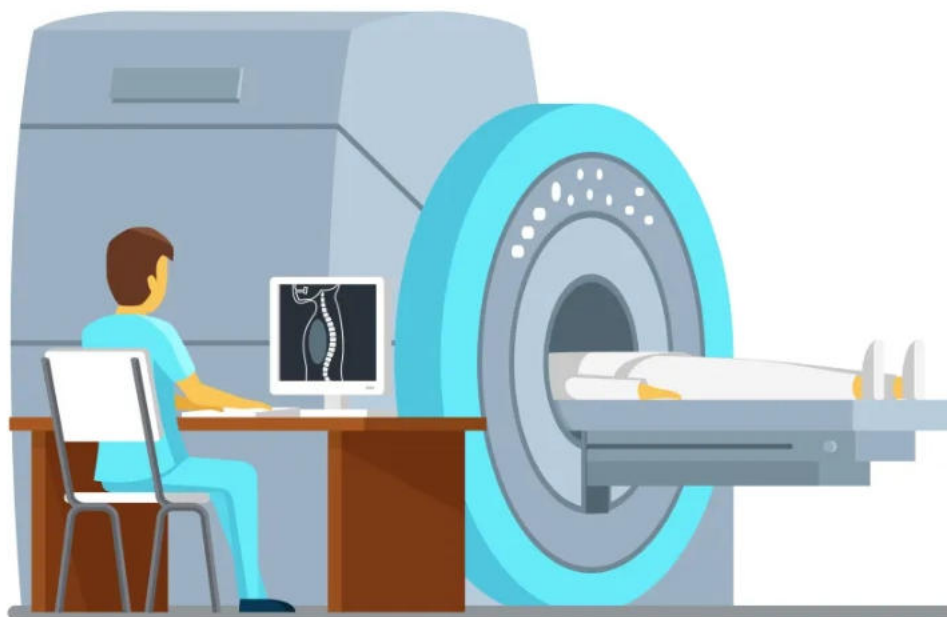


3. Hybrid Computers

Hybrid computers combine the features of analog and digital computers. Analog computers perform measurements, while digital computers control their operations.

Applications:

- Hospitals (e.g., for monitoring patients)
- Meteorology departments



Types of Digital Computers by Purpose

1. Special Purpose Computers

These computers are designed to perform a single specific task. Their programs or instructions are permanently stored in the machine.

Examples:

- Military applications
- Billing counters in malls



2. General Purpose Computers

These computers can solve a wide variety of problems by accepting different sets of inputs.

Examples:

- Banking systems
- Sales analysis systems



Types of Digital Computers by Size and Performance

1. Embedded Computers

Embedded computers are integrated within the circuitry of appliances and perform specific tasks.

Examples:

- Televisions
- Washing machines
- Refrigerators
- Motorbikes

2. Microcomputers

Microcomputers are single-user systems designed for basic operations like educational activities and gaming. They are used in homes, offices, shops, and schools.

Categories of Microcomputers:

- **Programmable Computers (PDA):** Used for sharing or exchanging information and as notepads or address books.
- **Laptops, Desktops, and Notebooks (Personal Computers):** Used for day-to-day personal tasks in business and at home.
- **Workstations:** Similar to personal computers but with higher memory and processing capabilities for industrial and scientific environments.



3. Mini Computers

Mini computers are more powerful than microcomputers and support multiple users. They are used for tasks requiring moderate computational power.



4. Mainframe Computers

Mainframe computers handle large volumes of data and support multiple users and processors. They are commonly used in large organizations.



5. Super Computers

Supercomputers perform highly complex computations in minimal time. They are the fastest, largest, and most expensive digital computers available.



Applications:

- Weather forecasting
- Nuclear science
- Aerodynamic modelling

The classification of computers helps us understand their functionality and areas of application. By distinguishing between analog, digital, and hybrid computers and further categorizing digital computers based on their purpose and performance, we can appreciate their role in various industries and everyday life.

EXERCISES

01: Choose the correct answer.

1. Which type of computer measures continuously varying physical quantities?

- a) Digital Computers
 - b) Analog Computers
 - c) Hybrid Computers
 - d) Super Computers
-

2. Which application is NOT typically associated with digital computers?

- a) Business analysis
 - b) Educational institutions
 - c) Monitoring patient health
 - d) Scientific research
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3. What is the primary feature of hybrid computers?

- a) Measuring physical quantities only
 - b) Combining analog and digital operations
 - c) Counting discrete data only
 - d) Performing simple arithmetic tasks
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4. What type of digital computer is embedded in appliances like refrigerators and televisions?

- a) Mini Computers
- b) Microcomputers
- c) Embedded Computers
- d) Workstations

5. Which type of computer is used for tasks requiring high memory and processing capabilities in scientific environments?

- a) Embedded Computers
 - b) Workstations
 - c) Mini Computers
 - d) Super Computers
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6. Which computers handle extremely complicated computations in the shortest possible time?

- a) Microcomputers
 - b) Super Computers
 - c) Mainframe Computers
 - d) General Purpose Computers
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7. What is an example of a special-purpose computer?

- a) Computer at a mall billing counter
 - b) Personal laptop
 - c) Banking system computer
 - d) Workstation used in industries
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8. Which category of microcomputers is designed for high computational ability in industrial environments?

- a) Laptops
 - b) Workstations
 - c) Personal Digital Assistants
 - d) Mini Computers
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9. What distinguishes general-purpose computers from special-purpose computers?

- a) They are embedded in appliances.
 - b) They can solve a wide variety of problems.
 - c) They operate only on analog data.
 - d) They perform only a single specific task.
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10. Which of the following is an application of supercomputers?

- a) Running educational programs
 - b) Performing weather forecasting
 - c) Operating a speedometer
 - d) Managing billing systems in malls
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02: Fill in the blanks.

1. _____ computers measure continuously varying quantities like pressure and temperature.
2. The programs in _____ purpose computers are permanently stored.
3. _____ computers are the fastest and most expensive digital computers.
4. _____ computers are designed for use within appliances like televisions and refrigerators.
5. A _____ computer is similar to a personal computer but has greater memory and processing power.
6. _____ computers are multi-user systems capable of handling large data volumes.

03: Select True or False for each statement.

	TRUE	FALSE
Analog computers operate on discrete data.	☐	☐
Digital computers are mainly used for counting.	☐	☐
Mainframe computers are designed for single-user tasks.	☐	☐
Supercomputers are best suited for simple arithmetic operations.	☐	☐
PDAs are a type of microcomputer.	☐	☐
Embedded computers are used in weather forecasting.	☐	☐

04: Answer the following questions briefly.

1. What are the three main categories of computers based on the data they handle?
2. Explain the difference between analog and digital computers.
3. What is the primary use of hybrid computers?
4. Provide examples of embedded computers.
5. How do general-purpose computers differ from special-purpose computers?
6. What are the main applications of supercomputers?

05: Provide detailed answers to the following questions.

1. Describe the classification of computers based on the type of data they handle, with examples.
2. Explain the differences between microcomputers, mini computers, and mainframe computers.
3. Discuss the characteristics and applications of supercomputers.
4. Provide a detailed comparison of special-purpose and general-purpose computers.

06: Match the computer type with its application.

S.No	A	S.No	B	C
1.	Analog Computers	(a)	Weather forecasting	
2.	Digital Computers	(b)	Educational institutions	
3.	Hybrid Computers	(c)	Refrigerators	
4.	Embedded Computers	(d)	Speedometer	
5.	Super Computers	(f)	Patient monitoring	

Lab Activity :1 Documenting Types of Computers



Objective: Identify different types of computers in your surroundings and document your observations.

1. Open MS Word and create a new document titled **"Types of Computers in My Surroundings."**
2. Create a table with the following columns:
 - Device Name
 - Type (Desktop, Laptop, Appliance, etc.)
 - Features (e.g., processor, screen size, portability, etc.)
3. Observe and classify at least five devices in your surroundings and fill in the table.
4. Add relevant images of the devices (use online sources or take photos) and format the document using headings, bullet points, and proper alignment.
5. Save the file as **"Device_Classification.docx."**

Lab Activity :2 Exploring Embedded Computers



Objective: Document the functionality of embedded computers using MS Word.

1. Open MS Word and create a new document titled

"Exploring Embedded Computers."

2. Write a brief introduction about embedded computers and their role in everyday devices.

3. Open a simple appliance (e.g., a remote-controlled toy, microwave, or calculator).

4. Create the following sections in the document:

- **Device Name:** Specify the appliance examined.
- **Identified Components:** Describe the embedded components observed.
- **Purpose and Functionality:** Write about how the embedded system controls the device.

5. Insert labelled diagrams or images of the appliance to illustrate your findings. Use text boxes and arrows for annotations.

6. Save the file as **"Embedded_Components.docx."**